

TATA STEEL AI CHALLENGE | ROUND 2

Maintenance Wizard

AI-Powered Industrial Equipment Decision Support

An enterprise-grade, condition-based predictive maintenance platform. Built with a 6-agent orchestration brain to perform diagnostics, forecast remaining useful life (RUL), and synchronize maintenance safety protocols (SOPs) with live spare parts inventory.

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Deployment link: [frontend-five-self-57.vercel.app](#)

1. Critical Downtime Costs

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Unplanned breakdowns in heavy industries (like steel production plants) cost upwards of **\$10,000+ per hour**.

Failure of critical machinery like Blast Furnace cooling pumps or Steel Melting Shop converters halts the entire production line, resulting in massive financial and production yield losses.

2. Invisible Anomalies

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Industrial plants capture millions of sensor parameters (temperature, pressure, vibration, current).

However, multivariate anomaly signatures remain **largely invisible** inside raw data streams. Maintenance teams struggle to diagnose complex machine decay before catastrophic failures manifest.

3. Fragmented Information

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Legacy plants suffer from severely fragmented information.

Standard Operating Procedures (SOPs), equipment manufacturer manuals, historical maintenance log books, and physical spare parts inventory records are isolated in paper formats or siloed databases, slowing down response teams.

Core Platform Offerings

- **Dynamic Plant Floor Mapping:** Active condition monitoring representing health indices across 25 machinery nodes in 6 critical plant environments.
- **AI Chat & Diagnostics:** Instant conversational agent to troubleshoot equipment faults and query historical maintenance records.
- **Spares Procurement integration:** Identifies asset degradation and automatically cross-checks active storage inventory to calculate financial exposure.

1. Condition-Based Prediction

Calculates asset health curves and Remaining Useful Life (RUL) using Isolation Forest models, moving away from calendar scheduling.

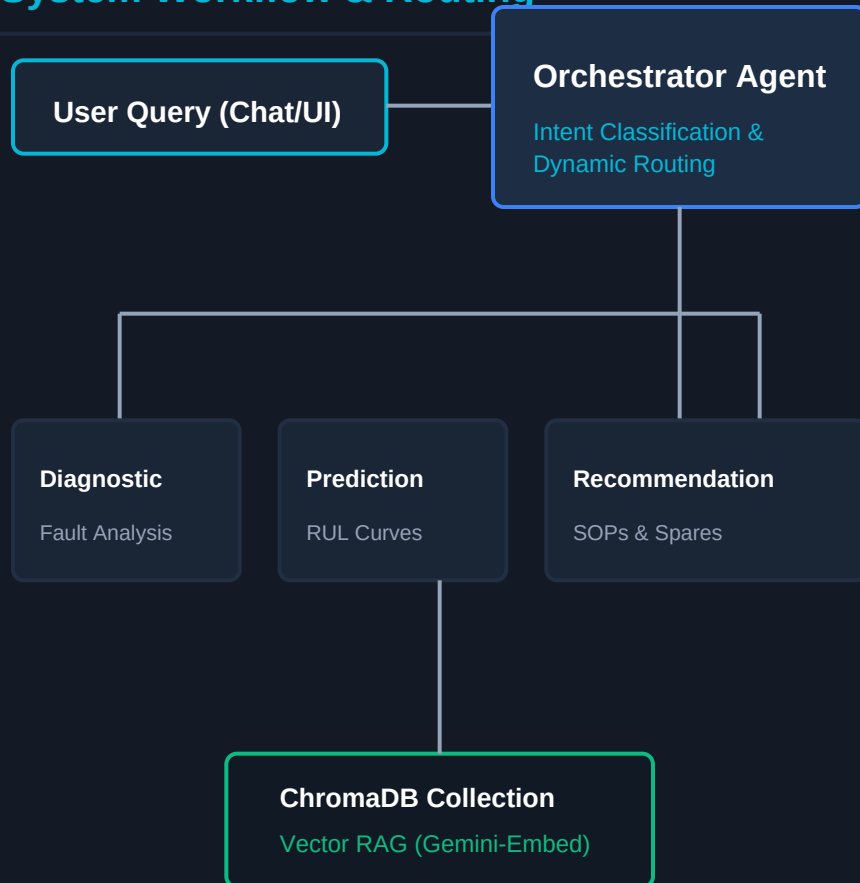
2. High-Performance RAG

Upload PDF manuals and SOPs directly. The vector engine extracts, indexes (ChromaDB), and answers domain questions within seconds.

3. Report Autopilot

Compiles structured executive summaries, incident reviews, and health reports instantly. Includes active feedback stores for AI training.

System Workflow & Routing



Platform Specifications

DEVELOPMENT STACK:

- **Frontend:** Next.js 14, React, TypeScript, TailwindCSS, Lucide Icons.
- **Backend:** FastAPI (Python 3.10), Uvicorn Reload server.
- **Vector Database:** ChromaDB (Persistent local storage).
- **ML Frameworks:** scikit-learn (Isolation Forest), NumPy, Pandas.

6-AGENT ORCHESTRATION CORE:

1. **Orchestrator:** Dynamic classification using Gemini/Llama.
2. **Diagnostic:** Sensor metrics threshold cross-analysis.
3. **Prediction:** Computes remaining useful life forecasts.
4. **Recommendation:** Safety compliance & replacement actions.
5. **Knowledge Agent:** Vectorized manual & SOP searches.
6. **Report Agent:** Autonomously compiles PDF reports.

Predictive Logistics & Procurement Integration

The system goes beyond technical predictions by directly integrating with the plant's **spare parts inventory database**.

When an anomaly detector flags a component (e.g., a bearing in a centrifugal pump) as degrading:

- The system instantly cross-references physical warehouse logs (`spare\_parts.json`).
- If replacements are **out of stock**, it computes the lead-time risk and exposes potential production downtime loss.
- Enables maintenance managers to initiate **1-Click procurement dispatch** right from the dashboard.

Downtime Prevention Calculator (ROI)

Demonstrates direct business value with a live financial yield tracker:

- **Production Loss Value:** Calculates downtime costs dynamically based on hourly throughput (e.g., converter vessel operating at 180 tonnes/hour).
- **Prevented Failure Analytics:** Displays a cumulative financial tracker showcasing total net savings achieved by resolving anomalies before critical failure.
- **Leaderboard:** Identifies and ranks machines with the highest degradation rates, focusing engineering resources on assets carrying the highest financial risk.

Dual-LLM Resiliency

Features a robust, enterprise-grade **unified LLM provider abstraction layer**.

The orchestrator operates on Groq (using Llama 3.3-70B) to provide sub-second chat response times.

In the event of Groq API rate limits, network timeouts, or service interruptions, the system **automatically falls back to Google Gemini 2.5** to maintain 100% platform availability.

Sub-2ms Sensor Cache

Industrial sensor logs contain massive datasets that are resource-intensive to parse during conversation.

We implemented a **custom in-memory caching database** for the multi-variate sensor logs.

This optimization dropped query latencies for diagnostic check commands from 200ms **to under 2ms**, ensuring instant AI response times.

Robust Vector Ingestion

Handles bulk upload of manuals, procedures, and failure files without bottlenecking:

- Implements **batch document ingestion** in packages of 20 vector chunks.
- Features an **exponential retry backoff algorithm** to smoothly handle embedding quota rate limits.
- Guarantees complete document indexing even under heavy network load.